

Mantain Waves (3D)

Background flow U along x -axis.

Assume horizontal form mode: $u, v \sim e^{i(kx + ly)}$

mom

$$\left\{ \begin{array}{l} -fv + Uv_x = -\phi_x \quad \rightarrow \quad ikU\hat{v} - f\hat{v} = -ik\hat{\phi} \\ fu + Uu_x = -\phi_y \quad \rightarrow \quad ikU\hat{u} + f\hat{u} = -il\hat{\phi} \\ Uw_x = -\phi_z + b \quad \rightarrow \quad ikU\hat{w} = -\hat{\phi}_z + \hat{b} \end{array} \right.$$

buoyancy

$$\left\{ \begin{array}{l} Ub_x + N^2 w = 0 \quad \rightarrow \quad ikU\hat{b} + N^2\hat{w} = 0 \end{array} \right.$$



CONT

$$\left\{ \begin{array}{l} u_x + v_y + w_z = 0 \quad \rightarrow \quad ik\hat{u} + il\hat{v} + \hat{w}_z = 0 \end{array} \right.$$

What is $\hat{w}$? $\hat{\phi}$?

- From buoyancy, $\hat{b} = i \frac{N^2}{\omega h} \hat{w}$

- With vertical momentum, $ikU\hat{w} = -\hat{\phi}_z + i \frac{N^2}{\omega h} \hat{w} \Rightarrow \hat{\phi}_z = i \left[\frac{N^2 - U^2 k^2}{\omega h} \right] \hat{w}$

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~~- With continuity, $ik\hat{u} + il\hat{v} + \hat{w}_z = 0$~~

- From horizontal mom., $f\hat{u} + ikU\hat{v} = i \frac{Uk(k^2 - l^2)}{f^2 - U^2 k^2} \hat{\phi}$

- With continuity, $ik\hat{u} + il\hat{v} = -\hat{w}_z \Rightarrow \hat{w}_z = i \left[\frac{Uk(k^2 - l^2)}{U^2 k^2 - f^2} \right] \hat{\phi}$

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We find

$$\hat{w}_z + \frac{(k^2 - l^2)(N^2 - U^2 k^2)}{U^2 k^2 - f^2} \hat{w} = 0$$

$$\hat{\omega}_{2t} \rightarrow m^1 \hat{\omega} = 0$$

$$w: k \quad \left[m^2 = (k^1 - l^1) \left[\frac{N^1 - U^1 k^1}{U^1 k^1 - f^1} \right] \right] \quad \text{For 3D, rotating non-hydrodynamic}$$

$$m^1 = (k^1 - l^1) \left[\frac{N^1}{U^1 k^1 - f^1} \right] \quad \text{For 3D, rotating hydro}$$

$$f=0, \quad m^1 = (k^1 - l^1) \left[\frac{N^1}{U^1 k^1} - 1 \right] \quad \text{For 3D, non-rotating non-hydro}$$

$$m^1 = (k^1 - l^1) \left[\frac{N^1}{U^1 k^1} \right] \quad \text{For 3D, non-rotating hydro}$$

$$l=0, \quad \left[m^2 = k^1 \left[\frac{N^1 - U^1 k^1}{U^1 k^1 - f^1} \right] \right] \quad \text{For 2D, rotating non-hydro}$$

$$m^1 = k^1 \left[\frac{N^1}{U^1 k^1 - f^1} \right]$$

$$m^1 = k^1 \left[\frac{N^1}{U^1 k^1} - 1 \right]$$

$$m^1 = \frac{N^1}{U^1}$$

Now with solution to $\hat{w}_{zz} + m^2 \hat{w} = 0$,

$$\hat{w}(t) \sim e^{\pm imz}$$

$$\begin{pmatrix} ikU & -f & 0 & 0 & ik \\ f & ikU & 0 & 0 & il \\ 0 & 0 & ikU & -1 & im \\ 0 & 0 & N^2 & ikU & 0 \\ ik & il & im & 0 & 0 \end{pmatrix} \begin{pmatrix} \hat{u} \\ \hat{v} \\ \hat{w} \\ \hat{\beta} \\ \hat{\phi} \end{pmatrix}$$

Horizontal component

$$\overset{A}{\begin{pmatrix} i\hbar U & -f \\ f & i\hbar U \end{pmatrix}} \begin{pmatrix} \hat{U} \\ \hat{N} \end{pmatrix} = \begin{pmatrix} -ik\hat{\phi} \\ -il\hat{\phi} \end{pmatrix} \Rightarrow A \begin{pmatrix} \hat{U} \\ \hat{N} \end{pmatrix} = \begin{pmatrix} -ik\hat{\phi} \\ -il\hat{\phi} \end{pmatrix}$$

$$\text{Det}(A) = \text{Det}(A) ((i\hbar U)^2 - f^2) = f^2 - U^2 \hbar^2$$

$$A^{-1} = \frac{1}{\text{Det}(A)} \begin{pmatrix} i\hbar U & f \\ -f & i\hbar U \end{pmatrix} = \frac{1}{f^2 - U^2 \hbar^2} \begin{pmatrix} i\hbar U & f \\ -f & i\hbar U \end{pmatrix}$$

$$\begin{pmatrix} \hat{U} \\ \hat{N} \end{pmatrix} = \begin{pmatrix} i\hbar U & f \\ -f & i\hbar U \end{pmatrix} \begin{pmatrix} -ik\hat{\phi} \\ -il\hat{\phi} \end{pmatrix} \frac{1}{f^2 - U^2 \hbar^2} \Rightarrow \begin{pmatrix} \hat{U} \\ \hat{N} \end{pmatrix} = A^{-1} \begin{pmatrix} -ik\hat{\phi} \\ -il\hat{\phi} \end{pmatrix}$$

$$\boxed{\begin{aligned} \hat{U} &= \frac{U\hbar^2 - if}{f^2 - U^2 \hbar^2} \hat{\phi} \\ \hat{N} &= \frac{U\hbar l + ikf}{f^2 - U^2 \hbar^2} \hat{\phi} \end{aligned}}$$

Vertical component

$$\hat{W} = - \frac{\hbar \hat{U} + l \hat{N}}{m}$$

$$\hat{b} = i\hbar U \hat{W} + im \hat{\phi}$$

From continuity

$$\boxed{\hat{W} = \frac{U\hbar (k^2 - l^2)}{m(U^2 k^2 - f^2)} \hat{\phi}}$$

$$\boxed{\hat{b} = i \left[m + \frac{U^2 \hbar^2 (k^2 - l^2)}{m(U^2 k^2 - f^2)} \right] \hat{\phi}}$$

From boundary

$$\hookrightarrow \hat{W} = \frac{m U \hbar}{N^2 - U^2 \hbar^2} \hat{\phi}$$

$$\hookrightarrow \hat{b} = \frac{i N^2}{U \hbar} \hat{W}$$

Boundary conditions

Lower (the mountain): $w = \frac{\partial h}{\partial t} + U \frac{\partial h}{\partial x} + V \frac{\partial h}{\partial y}$ for stationary mountain assuming boundary flow in x only ($V=0$)

At $z = h(x, y)$, $w(x, y, h(x, y), t) = \cancel{\frac{\partial h}{\partial t}} + U h_x + 0$

Linearized at $z=0$, $w(x, y, 0) = \cancel{\frac{\partial h}{\partial t}} + U h_x$

$$\hat{w}(k, l, 0) = ikU \hat{h}(k, l)$$

$\downarrow$
 $z=0$ (not $h=0$)

Upper: - waves must go upward only for propagating modes
oscillatory, $m > 0$, $w(z) \sim e^{\pm imz}$

- waves must decay with height for evanescent modes
exponential, $m^2 < 0$, $w(z) \sim e^{-|m|z}$ when $m = i\alpha$
 $\hookrightarrow$ keep only decaying branch

~~for stationary mountain~~

$$\hat{\phi}(k, l, 0) = \frac{N^2 - U^2 k^2}{mUk} \hat{w}(k, l, 0)$$

~~for stationary mountain~~

~~Vertical component~~

$$\vec{w} = \underbrace{h\vec{v} + \frac{p\vec{r}}{m}}_{\vec{w}_0}$$

$$\vec{\phi}_0 = m \left(\frac{h\vec{v}}{m} + \frac{p\vec{r}}{m} \right) = \vec{w}_0$$

$$\vec{\phi}_0 = \vec{w}_0$$

Angularities:

$$h = 0$$

$$U^2 h^2 = f^2$$

$$U^2 h^2 = N^2$$

$$h = 0$$

$$h^2 = l^2 = 0$$